ALFIDO TECH – DATA ANALYTICS INTERNSHIP

TASK 1: CUSTOMER BEHAVIOR ANALYSIS

Submitted by: MIRSAJI S | **Tools Used:** Python, Pandas, NumPy, Matplotlib, Seaborn | **Environment:** Google Colab

PAGE 1: PROJECT OVERVIEW

1. Objective

The objective of this project is to analyze customer purchasing behavior and identify useful patterns in sales, payment preferences, age-group spending, and customer value. RFM (Recency, Frequency, and Monetary) analysis is used to understand customer segments and support better business decisions.

2. Dataset Overview

The project uses an e-commerce customer dataset containing purchase records, customer details, product categories, purchase amounts, payment methods, and purchase dates. The analysis includes data inspection, data cleaning, feature preparation, visualization, and customer segmentation.

3. Executive Summary

This project analyzes customer purchasing behavior using Python and data analysis libraries. The analysis examines product category sales, payment preferences, spending across age groups, and RFM-based customer segments.

The results reported in the analysis show that Books generated the highest sales among the product categories, Credit Card was the most commonly used payment method, and the 56+ age group had the highest total spending. The report also identifies High Value, Loyal, Potential, and At Risk customer segments using RFM analysis.

These findings can help businesses improve customer retention, create targeted marketing campaigns, and make better product-promotion decisions.

PAGE 2: ANALYSIS AND VISUALIZATIONS

4. Product Category Sales Analysis



Figure 1: Total Sales Revenue by Product Category.

The analysis compares total sales across Books, Electronics, Clothing, Beauty, and Home. Books recorded the highest reported sales, at approximately Rs. 204.9 million, while Home recorded the lowest, at approximately Rs. 135.2 million. **Key finding:** Sales performance differs across product categories, with Books contributing the highest reported revenue.

5. Payment Method Analysis

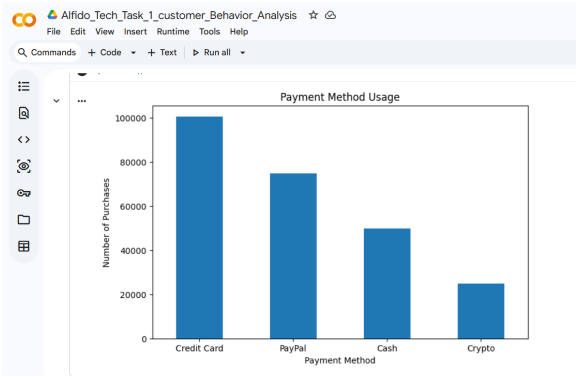


Figure 2: Distribution of Customer Payment Methods.

The analysis compares the use of Credit Card, E-Wallet, Debit Card, and Cash. Credit Card accounted for 41%, E-Wallet for 27%, Debit Card for 18%, and Cash for 14%. **Key finding:** Credit Card was the most frequently used payment method in the analyzed dataset.

PAGE 3: RECOMMENDATIONS AND CONCLUSION

6. Age Group Spending Analysis



Figure 3: Total Spending by Age Group.

Customer spending is compared across age groups (18–25, 26–35, 36–45, 46–55, 56+). The 56+ group recorded the highest total spending (~Rs. 78 million). **Key finding:** The 56+ age group contributed the highest total spending among all age groups analyzed.

7. RFM Customer Segmentation

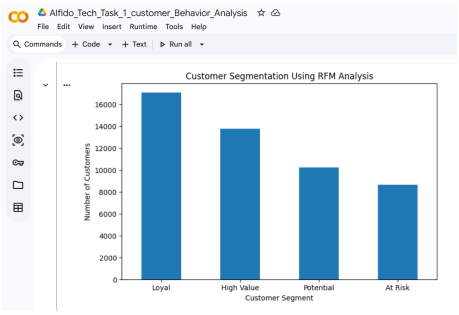


Figure 4: Customer Segmentation Using RFM Analysis.

RFM analysis evaluates customers by Recency, Frequency, and Monetary value into High Value, Loyal, Potential, and At Risk segments. **Key finding:** RFM segmentation distinguishes valuable and inactive customers.

8. Three Actionable Insights and Recommendations

- **Customer Retention:** Give loyalty rewards to High Value customers and targeted offers to At Risk customers.
- **Cross-Selling:** Recommend related products and create bundles based on the strong Books sales.
- **Targeted Marketing:** Use age-group spending and payment preferences to plan suitable campaigns.

9. Overall Outcome

The analysis demonstrates how customer data can be used to understand purchasing patterns, compare product categories, examine payment preferences, evaluate age-group spending, and segment customers through RFM analysis. The findings provide a basis for customer retention strategies, targeted promotions, and further business analysis.

10. Conclusion

Customer behavior analysis helps businesses make informed decisions using transaction data. Combining descriptive analysis, visualizations, and RFM segmentation provides useful insights into customer value and purchasing behavior.

11. Notebook Link

Task 1 – Customer Behavior Analysis Notebook:
https://github.com/mirsaji01-star/Alfido_Tech_Data_Analytics/blob/main/Alfido_Tech_Task_1_customer_Behavior_Analysis.ipynb

End of Report